

PARTS LIST

Semiconductors:

- 7805, 5V regulator
- T1 - BC549 npn transistor
- D1, D2 - 1N4148 rectifier diode
- D3, D4 - 1N4001 rectifier diode
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon unless stated otherwise):

- R1, R2, R4 - 1-kilo-ohm
- R3 - 100-kilo-ohm
- R5 - 100-ohm, 0.5W
- R6 - 3-kilo-ohm

Capacitors:

- C1 - 1 μ F, 400V electrolytic
- C2, C4, C7, C9 - 0.33 μ F ceramic disk
- C3 - 1 μ F, 63V electrolytic
- C5 - 33pF ceramic disk
- C6 - 220 μ F, 16V electrolytic
- C8 - 470 μ F, 35V electrolytic

Miscellaneous:

- Ant - Longer than 2 metres
- CON1, CON2, CON3 - Two-pin connector

long and preferably mounted outside the building at least half a metre away from the walls in order to have noticeable improvement in reception. If you keep the antenna near walls or electrical cable installation, there may be reception issues and the antenna may capture too much noise. When there

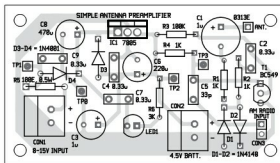


Fig. 3: Component layout for the PCB

is possibility of lighting storms in the area, the antenna must be disconnected from the preamplifier.

The preamplifier can work off a 4.5V dry battery connected to connector CON2. Use of batteries in the range of 3V to 15V is recommended because the mains power supply may introduce some noise.

The preamplifier does not use expensive components. It will work immediately after proper assembly is done and show improvement in the reception of stations with low level of AM signal. The improvement depends

on the location and length of the antenna. If you use the antenna in a room, you should put it on a window in order to increase the useful signal and reduce noise from other installations in the building.

To test the circuit for proper functioning, check the input supply at test point TP1 with respect to

TP0. The 5V supply to the preamplifier can be checked at TP2. Test the gain of the circuit by giving some input signal, using ordinary sinusoidal signal generator, to the preamplifier and checking the output at TP3, using an oscilloscope. For a 10mV, 1MHz signal at the input (antenna), you should get around 100 mV at TP3. ●

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